

Abstract

This project develops a machine-learning baseline for classifying oral images into cancer and non-cancer classes. The supplied dataset contains 131 images. Images were converted to grayscale, resized to 128×128 pixels, and represented using Histogram of Oriented Gradients (HOG) descriptors. Logistic Regression was used as the classifier. Using an 80/20 stratified split, the model achieved 81.48% accuracy, 81.75% precision, 81.48% recall, and 80.34% F1-score. Cancer recall was approximately 94%, while non-cancer recall was approximately 56%.

1 Problem Statement and Dataset

The objective is to build an end-to-end binary image-classification pipeline that predicts whether an oral image belongs to the *cancer* or *non-cancer* class.

The dataset was supplied as a ZIP archive and contains two class directories. Directory names were used as target labels.

Table 1: Dataset Summary

Property	Value
Total images	131
Cancer	87
Non-cancer	44
Classes	2
Target	label
Input size	128×128
HOG features/image	8,100
Train/Test	80%/20%

The dataset contains 66.4% cancer and 33.6% non-cancer images. Stratification was used during splitting to preserve class proportions.

2 Data Analysis

The dataset was checked for missing values, duplicate records, and image readability. No missing values were present in the constructed dataframe.

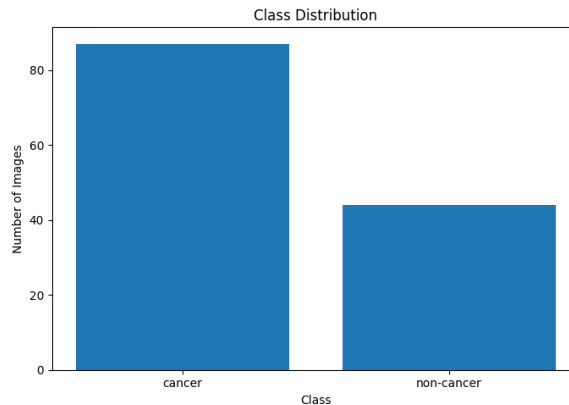


Figure 1: Class distribution.

Average image statistics were:

Class	Width	Height	Brightness
Cancer	769.09	601.44	130.67
Non-cancer	855.09	655.55	126.38

Correlation analysis showed strong width-height correlation ($r = 0.884$), while width-brightness ($r = 0.060$) and height-brightness ($r = 0.003$) were negligible. These are image-level properties and are not clinical correlations.

3 Preprocessing and Feature Extraction

The processing pipeline was:

Image $\rightarrow$ Grayscale $\rightarrow 128 \times 128 \rightarrow$ HOG $\rightarrow 8,100$ features.

HOG captures local intensity gradients and edge orientations and converts each image into numerical descriptors suitable for classical machine-learning algorithms.

The final feature matrix was:

$$X = (131, 8100), \quad y = (131,).$$

Labels were encoded numerically. An 80/20 stratified train-test split with `random_state=42` was used.

4 Model Selection and Training

Logistic Regression was selected as the baseline because it is efficient and appropriate for binary classification using high-dimensional numerical features. A linear **SVM** was considered as an alternative because it can perform well with HOG-based representations.

The model was trained only on the training set and evaluated on unseen test images.

5 Results

The test set contained 27 images.

Table 2: Test-set Performance

Metric	Score
Accuracy	81.48%
Precision	81.75%
Recall	81.48%
F1-score	80.34%

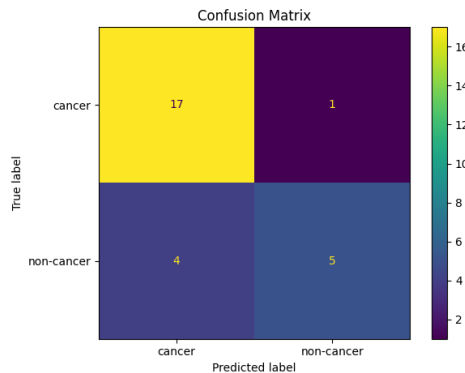


Figure 2: Confusion matrix.

The model correctly classified 22/27 test images. Cancer recall was $17/18 \approx 94\%$, while non-cancer recall was $5/9 \approx 56\%$.

6 Explanation of Results

Is the model performing well? The model provides a reasonable baseline with 81.48% accuracy and 80.34% F1-score. Its cancer recall of approximately 94% is strong on this test split, but the 56% non-cancer recall indicates substantial false positives. Therefore, the model is promising experimentally.

Major reasons for errors:

- Only 131 images are available; the test set contains only 27 images.
- The classes are imbalanced (87 cancer vs. 44 non-cancer).
- Images vary in resolution, lighting, background, and acquisition conditions.
- HOG captures general visual patterns rather than medically validated biomarkers.

Which features influence results? The model uses 8,100 HOG features representing local gradient and edge patterns. Logistic Regression coefficients can be examined to identify influential HOG dimensions. However, individual HOG indices should not be interpreted as clinical biomarkers. Image width, height, and brightness are descriptive statistics rather than the model's primary predictive features.

How would the model be improved? The first priority would be a larger and more diverse dataset. I would then use stratified cross-validation, investigate class-weighting and controlled augmentation, compare SVM and other classifiers, improve image

standardization, and investigate transfer learning using pretrained CNNs such as ResNet, MobileNet, or EfficientNet. Finally, evaluation on an independent external dataset would be required for assessing generalization.

7 Final Result

The complete demonstration follows the required pipeline:

Dataset → Preprocessing → Training → Prediction → Results

1. **Dataset:** Load the cancer and non-cancer image directories and create labelled samples.
2. **Preprocessing:** Convert images to grayscale and resize them to 128×128 pixels.
3. **Feature extraction:** Generate 8,100 HOG features for each image.
4. **Training:** Perform an 80/20 stratified split and train Logistic Regression.
5. **Prediction:** Capture or upload a single unseen image and apply the same preprocessing and HOG extraction pipeline.
6. **Results:** Generate the predicted class and model confidence and evaluate the model using classification metrics.

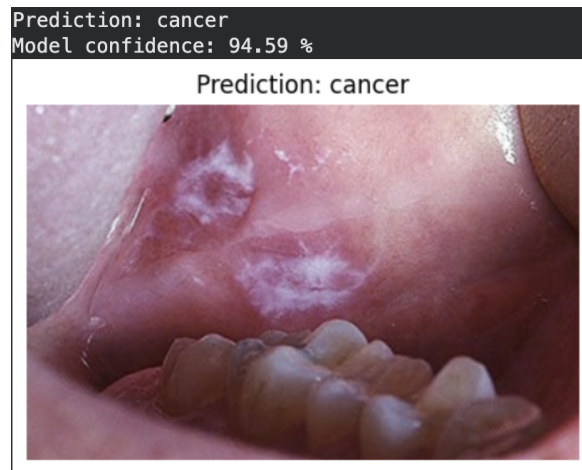


Figure 3: Example single-image prediction using the trained model.

The prediction shown is the output of the machine-learning model and confidence score.

8 Limitations and Conclusion

The main limitation is the small dataset size and the resulting small test set. Performance can therefore vary considerably with the train-test split. HOG also represents general image structure rather than clinically validated features.

Overall, the project demonstrates a complete reproducible pipeline for oral image classification. The Logistic Regression baseline achieved 81.48% accuracy and approximately 94% cancer recall on the supplied test split. The results are encouraging for an educational prototype but require substantially more data, stronger validation, and external testing before any real-world healthcare application could be considered.

Code and Reproducibility

The complete implementation, including dataset loading, preprocessing, HOG feature extraction, model training, evaluation, and single-image prediction, is available:

[Open Notebook](#)

The notebook contains the complete executable workflow used to obtain the reported results.